

LEAD AGENT

Project Requirement Document

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1. Executive Summary

Lead Agent is an AI-powered inbound lead qualification and research automation platform developed as a **Capstone Individual Project** over **5 weeks**. It captures prospect submissions through a React contact form, runs a durable multi-phase AI pipeline in the background, and surfaces a curated email draft to a Slack channel for one-click human approval — all within minutes of a form being submitted.

The core intelligence engine classifies every inbound lead into one of **four categories**: **QUALIFIED** (sales-ready), **FOLLOW_UP** (nurture track), **UNQUALIFIED** (no fit), or **SUPPORT** (customer service routing). Classification combines deterministic keyword rules with a structured LLM call ensuring both speed and accuracy.

Built on **Next.js 16**, **Vercel AI SDK**, **Workflow DevKit**, **Exa.ai**, and **Slack Bolt**, the platform is hosted on the **Vercel AI Cloud** and is designed as a configurable reference architecture any B2B sales team can adopt and extend.

2. Product Vision & Goals

Vision Statement

"Enable every sales team — regardless of size — to respond to inbound leads with the speed of automation and the precision of expert human judgment."

Strategic Goals

#	Goal	Rationale
G-1	Reduce lead response time from hours to minutes	Faster response directly increases conversion
G-2	Achieve >90% qualification accuracy vs manual review	Eliminate wasted time on unfit leads
G-3	Provide enriched intelligence on every lead automatically	Give sales reps context before any conversation
G-4	Maintain mandatory human approval before any outbound email	Prevent unsupervised AI emails reaching prospects
G-5	Configurable ICP rules for any B2B organisation	Make the platform reusable across verticals

3. Problem Statement

The Current Gap in B2B Sales

B2B sales teams receive inbound leads through contact forms containing only basic information — name, email, company, and a short message. Qualifying each lead manually forces a sales representative to:

- Research the company on LinkedIn, Crunchbase, and Google News

- Verify the person's title and purchasing authority
- Assess ICP fit — company size, industry, geography, tech stack
- Draft a personalised, context-aware email response
- Obtain manager sign-off before sending

This process consumes **2–8 hours per lead** and quality is entirely rep-dependent — inconsistent, unscalable, and slow. High-value prospects frequently go cold because competitors respond faster.

Business Impact

Problem	Business Cost
Slow lead response	High-value leads go cold — competitors respond faster
Manual qualification inconsistency	Good leads misrouted; unfit leads waste rep time
No shared research standard	Every rep researches differently — no institutional memory
Volume spikes unmanageable	Manual process breaks under high inbound volume

4. Target Users & Personas

Persona 1 — Sales Development Representative (SDR)

The primary user. Receives Slack notifications with pre-researched lead intelligence, reviews the AI-generated email draft, and clicks Approve or Reject. Gains an estimated **2+ hours saved per lead** with no manual googling required.

Persona 2 — Sales Manager / VP of Sales

Monitors pipeline quality and SDR performance through the Slack audit trail. Responsible for keeping ICP rules accurate each quarter and reviewing override rates to catch systematic qualification errors.

Persona 3 — Growth / Marketing Engineer

Deploys and configures the platform. Updates ICP rules in `qualification-rules.ts`, adjusts AI prompts in `services.ts`, and implements org-specific integrations for CRM, email delivery, and knowledge base lookup.

Persona 4 — Inbound Lead (Prospect)

Submits the contact form on the company website. Receives a timely, personalised email that directly references their company, use case, and current technology stack rather than a generic autoresponder.

5. Scope & Feature Requirements

In Scope — Version 1.0

Feature	Priority	Status
Lead capture form (name, email, phone, company, message)	P0	Complete
Immediate HTTP 200 response after form submit	P0	Complete
Durable background workflow (Workflow DevKit)	P0	Complete
Phase 1: Background verification — email, company, decision-maker, tech, financial, risk	P0	Complete
Phase 2: AI deep research agent via Exa.ai search + fetch	P0	Complete
Phase 3: Compiled research report (1,500–3,000 words)	P0	Complete
Lead qualification into 4 categories (QUALIFIED / FOLLOW_UP / UNQUALIFIED / SUPPORT)	P0	Complete
AI-generated personalised email draft per qualification category	P1	Complete
Slack notification with Approve / Reject action buttons	P0	Complete
Email delivery on Approve action	P1	Pending stub
Configurable ICP rules file (qualification-rules.ts)	P1	Complete

Out of Scope — Version 1.0

- CRM write-back (Salesforce, HubSpot) — planned v1.1
- SMS / WhatsApp notifications — Slack-only in v1.0
- Multi-language support — English-only in v1.0
- Lead analytics dashboard — logs + Vercel monitoring in v1.0
- A/B email template testing
- Lead deduplication across multiple submissions

6. Functional Requirements

FR-01 Lead Capture Form

ID	Requirement	Priority
FR-01-1	Capture: name (required), email (required), phone (optional), company (optional), message (10–500 chars)	P0
FR-01-2	Validate all fields with Zod schema — client-side and server-side	P0
FR-01-3	Return HTTP 200 and trigger background workflow immediately on valid submission	P0
FR-01-4	Return HTTP 422 with field-level error messages on invalid data	P0

FR-02 Background Verification (Phase 1)

ID	Requirement	Priority
FR-02-1	Validate email domain — check if corporate, real, and reputable	P0
FR-02-2	Research company — retrieve industry, employee count, location, LinkedIn URL	P0
FR-02-3	Verify decision-maker title level: C-Suite / VP / Manager / IC	P0
FR-02-4	Analyse tech stack compatibility with the product	P0
FR-02-5	Check financial health — funding news, growth signals, red flags	P1
FR-02-6	Assess overall risk level: low / medium / high with contributing factors	P1

FR-03 AI Deep Research Agent (Phase 2)

ID	Requirement	Priority
FR-03-1	Use Exa.ai search() to find recent news about the company and individual	P0
FR-03-2	Use Exa.ai fetchUrl() to read and summarise specific web pages	P0
FR-03-3	Call crmSearch() to surface any existing CRM history for the lead	P1
FR-03-4	Call techStackAnalysis() for deeper compatibility assessment	P1
FR-03-5	Call queryKnowledgeBase() for internal company intelligence	P2
FR-03-6	Minimum research output: 500 words	P1

FR-04 The Four Qualification Categories

Every inbound lead is classified into exactly one of the following four categories. The system first applies fast-path deterministic keyword rules, then falls back to a structured LLM call using **generateObject** with a constrained Zod enum schema (`QUALIFIED | FOLLOW_UP | UNQUALIFIED | SUPPORT`).

QUALIFIED Sales Ready

Qualifying signals:

- Clear business need that directly matches the product offering
- Company size within ICP range (20–5,000 employees by default)
- Decision-maker present — CTO, VP Engineering, Director of Engineering
- Urgency or immediate buying intent expressed in the message
- High or medium tech-stack compatibility detected
- Target industry and geography confirmed

Email action: Excited, personalised, includes CTA to book a demo. Sent to Slack for SDR approval. On approval — email sent. Priority outreach.

FOLLOW_UP Nurture Track

Qualifying signals:

- Genuine interest expressed but budget cycle is next quarter
- Company is exploring options rather than ready to commit
- Below ICP minimum size but shows strong growth trajectory
- Missing key information — needs a clarifying question first
- Interest in the product but on a seasonal or deferred timeline

Email action: Nurturing tone 'happy to keep you in the loop'. Approved via Slack. Lead added to drip sequence with 2–4 week re-engagement.

UNQUALIFIED ✕ Not a Fit

Qualifying signals:

- Excluded industry detected: Insurance, Government, Non-profit, Healthcare
- Keywords indicating personal/non-commercial use: freelancer, student, hobby project
- Company too small to meet minimum ICP size threshold
- Competitor company domain identified
- Use-case fundamentally outside the product scope

Email action: Optional polite decline (configurable) or silent archive. No SDR time spent. Lead logged and dismissed.

SUPPORT 🛠 Customer Service

Qualifying signals:

- Keywords detected: bug, error, broken, not working, can't login, account issue, refund
- Indicates an existing customer or user with a post-sale problem
- Not a new sales opportunity — requires support team routing

Email action: Support routing email drafted. Routed to #support Slack channel. SDR not involved.

FR-05 Email Generation

ID	Requirement	Priority
FR-05-1	Generate personalised email from lead data + research report + qualification category	P1
FR-05-2	Tone adapts per category: excited (QUALIFIED), nurturing (FOLLOW_UP), supportive (SUPPORT)	P1
FR-05-3	Email must reference specific company details and use-case from the research	P1

FR-06 Slack Human-in-the-Loop Approval

ID	Requirement	Priority
FR-06-1	Post structured Slack message to configured channel after every qualification	P0
FR-06-2	Message shows: category badge, 300-char research summary, 250-char email preview, action buttons	P0
FR-06-3	[☒ Approve] — sends email to prospect, posts confirmation in Slack thread	P0
FR-06-4	[☒ Reject] — cancels send, logs rejection reason in Slack	P0
FR-06-5	Webhook must acknowledge Slack within 3 seconds (hard platform limit)	P0

7. Non-Functional Requirements

Performance

Metric	Target
Form submit → Slack notification (full pipeline)	< 3 minutes at P95
/api/submit endpoint response time	< 500 ms
Slack action acknowledgement	< 3 seconds (hard Slack limit)

Reliability & Durability

Metric	Target
Workflow step durability	Steps are retryable on failure (Workflow DevKit guarantee)
System uptime	99.9% (Vercel SLA)
Research step failure	Graceful fallback to raw lead data; pipeline always continues

Security

Control	Detail
Slack request verification	Signing secret validated on every incoming webhook payload
Secret management	All API keys in Vercel environment variables — never in source code
Input sanitisation	Zod schema validates all form inputs before any processing begins
Rate limiting	Form endpoint recommended: max 10 requests / min / IP
Prompt injection	Message field length-capped at 500 chars; system prompt hardened

8. System Architecture Overview

Technology Stack

Layer	Technology	Purpose
Frontend	Next.js 16 (App Router)	Lead capture form and client-side UI
Backend API	Next.js Serverless API Routes	Form handler and Slack webhook endpoint
Workflow Engine	Workflow DevKit	Durable, retryable background pipeline
AI / LLM	Vercel AI SDK + AI Gateway	Qualification, email generation, research agent
Web Research	Exa.ai	Semantic search and URL content fetching
Human-in-Loop	Slack Bolt + Vercel Slack Adapter	Approval notification and action handling
Hosting	Vercel AI Cloud	Serverless deployment, auto-scaling
Validation	Zod	Schema validation for all data contracts
Styling	Tailwind CSS + shadcn/ui	Component library and design system

End-to-End Pipeline

```
[User] → Contact Form Submission
    ↓
[POST /api/submit] → HTTP 200 returned immediately
    ↓ (async trigger via Workflow DevKit)

Step 1 · stepResearch
├─ Phase 1: Background Verification
│   │ email domain · company data · decision-maker
│   │ tech stack · financial health · risk score
├─ Phase 2: AI Research Agent (Exa.ai tools)
│   │ search() · fetchUrl() · crmSearch() · queryKB()
    ↓
Step 2 · stepQualify
├─ Keyword rules → SUPPORT / UNQUALIFIED (fast-path)
├─ LLM generateObject → QUALIFIED / FOLLOW_UP / UNQUALIFIED
    ↓
Step 3 · writeEmail
├─ LLM generateText → personalised email draft
    ↓
Step 4 · stepHumanFeedback
├─ POST Slack message → [☒ Approve] [☒ Reject]
    ↓
[POST /api/slack] Slack webhook handler
├─ Approve → Send email → Slack confirmation
├─ Reject → Archive → Slack log entry
```

9. User Journeys & Workflows

Journey 1 — QUALIFIED Lead (Happy Path)

1. Prospect fills the contact form on the company website
2. System returns an immediate success confirmation
3. Background workflow launches within seconds
4. Phase 1: Company validated, decision-maker confirmed, low risk score
5. Phase 2: AI agent finds recent funding news via Exa.ai
6. Research report compiled — 1,500+ words across 8 sections
7. LLM classifies the lead as QUALIFIED with a written reason
8. Personalised email drafted referencing the company's specific context
9. Slack message sent: QUALIFIED badge, research preview, email preview, action buttons
10. SDR reviews and clicks [☒ Approve]
11. Email sent to prospect — Slack confirmation posted in thread

Journey 2 — FOLLOW_UP Lead

1. Prospect submits form expressing interest but mentions 'budget starts next quarter'
2. Full research pipeline completes
3. LLM detects timing mismatch — classification: FOLLOW_UP
4. Nurture email drafted with a 'stay in touch' tone
5. Slack notification sent with FOLLOW_UP badge and email preview
6. SDR approves — nurture email sent — lead added to re-engagement tracker

Journey 3 — UNQUALIFIED Lead

1. Submission from a student inquiring about a homework project
2. Keyword 'student' detected — fast-path classification: UNQUALIFIED
3. No LLM call needed — pipeline short-circuits immediately
4. Slack notification sent with reason: 'Personal/academic use detected'
5. SDR dismisses — no email sent — lead archived

Journey 4 — SUPPORT Request

1. Existing customer submits form: 'My account login is broken'
2. Keyword engine detects 'login' and 'broken' immediately
3. Category set to SUPPORT — sales pipeline bypassed
4. Support routing email drafted
5. Slack notification sent to #support-team channel
6. Support team handles — SDR not involved

10. Key Performance Indicators (KPIs)

10.1 Operational KPIs

KPI	Definition	Target	How Measured
Pipeline Throughput Time	Form submit → Slack notification	< 3 min P95	Workflow step timestamps
Step Success Rate	% steps completing without error	> 99%	Vercel error logs
Webhook Reliability	% Slack actions processed successfully	> 99.5%	Slack delivery logs
Research Depth	Average word count of compiled report	> 1,500 words	Log analysis
System Uptime	Availability of form + API endpoints	> 99.9%	Vercel uptime monitor

10.2 Sales Effectiveness KPIs

KPI	Definition	Target	How Measured
Lead Response Time	Form submit → email sent or approved	< 5 min avg	Timestamp logging
Qualification Accuracy	AI category agreed by SDR without override	> 85%	Slack approval rate
Email Approval Rate	AI email approved without editing	> 70%	Slack action logs
QUALIFIED Conversion	QUALIFIED leads → meeting or demo booked	> 20%	CRM (manual v1.0)
FOLLOW_UP Re-engagement	FOLLOW_UP → convert within 90 days	> 15%	CRM (manual v1.0)
False Positive Rate	UNQUALIFIED leads that were actually good fits	< 5%	SDR override tracking
False Negative Rate	QUALIFIED leads classified incorrectly lower	< 10%	SDR override tracking

10.3 Adoption & Efficiency KPIs

KPI	Definition	Target
Daily Leads Processed	Leads through full pipeline per day	Baseline Week 1
SDR Time Saved	Hours saved vs. manual per lead	> 2 hours / lead
Slack Response Time	Notification → SDR action (business hours)	< 30 minutes
Weekly Volume Capacity	Leads handled / week without degradation	500+ leads / week

11. Risks & Mitigation

ID	Risk	Likelihood	Impact	Level	Mitigation
R-01	Exa.ai unavailable or returns sparse data	Medium	High	HIGH	Retries with backoff; fallback to raw lead data; pipeline continues regardless
R-02	LLM returns wrong qualification category	Medium	High	HIGH	Human Slack approval is the final gate; monitor and track override rates weekly
R-03	Slack webhook timeout (> 3 seconds)	Low	Medium	MEDIUM	Async acknowledgment pattern: HTTP 200 immediately, process action in background
R-04	PII data exposed in server logs	Low	Critical	HIGH	Mask all PII fields in logs; use Vercel log drain with data masking enabled
R-05	AI generates inappropriate email draft	Medium	High	HIGH	Human approval mandatory — no email is ever sent without explicit SDR approval
R-06	Workflow DevKit service degradation	Low	High	MEDIUM	Step timeouts configured; alerts on step failures; monitor DevKit health dashboard
R-07	Prompt injection via message field	Low	High	MEDIUM	Zod max-length at 500 chars; system prompt hardened against injection patterns
R-08	ICP rules not updated to match real business	High	High	HIGH	Quarterly ICP config review scheduled; zero-code config file lowers barrier
R-09	Exa.ai rate limits under peak lead volume	Medium	Medium	MEDIUM	Request queuing layer; per-lead throttling; cache company lookups
R-10	Slack bot token or signing secret leaked	Low	Critical	HIGH	Rotate tokens quarterly; enable secret scanning in CI/CD pipeline

12. Dependencies

External Service Dependencies

Service	Purpose	Criticality	Fallback
Vercel AI Gateway	LLM inference for all AI calls	Critical	No fallback — workflow pauses if unavailable
Exa.ai	Web search and URL content fetching	High	Fallback to raw lead data only
Slack API	Human-in-loop notifications and actions	High	Workflow retries; notification queued
Vercel Platform	Hosting and serverless execution	Critical	Covered by Vercel platform SLA
Workflow DevKit	Durable background workflow execution	Critical	Entire pipeline depends on it

Required Environment Variables

Variable	Purpose	Required
AI_GATEWAY_API_KEY	Vercel AI Gateway authentication	Yes
SLACK_BOT_TOKEN	Slack API bot authentication	Yes
SLACK_SIGNING_SECRET	Slack webhook signature verification	Yes
SLACK_CHANNEL_ID	Target Slack channel for lead notifications	Yes
EXA_API_KEY	Exa.ai web research API access	Yes

13. Roadblocks & Challenges

Active Roadblocks

ID	Roadblock	Impact	Resolution Path
RB-01	lib/services.ts contains stub functions (crmSearch, sendEmail, queryKnowledgeBase)	Pipeline incomplete until stubs are implemented	Each adopting org implements per their CRM and email provider
RB-02	No CRM integration in v1.0	Conversion tracking is manual	CRM write-back (Salesforce / HubSpot) planned for v1.1
RB-03	Slack interactivity URL must point to production domain	Cannot test HIL locally without a tunnel	Use ngrok locally; document in onboarding guide
RB-04	ICP rules contain placeholder competitor and industry names	Incorrect qualifications until config is updated	Mandatory ICP config review step in the deployment checklist
RB-05	No persistent lead storage in v1.0	No historical queries or analytics dashboard	Add Vercel Postgres / Neon database in v1.1

Technical Challenges Encountered (5-Week Build)

Slack's 3-Second Hard Timeout

Slack requires an HTTP acknowledgement within 3 seconds of any interactive action, but lead processing takes significantly longer. Resolved using an async acknowledgement pattern: return HTTP 200 immediately, then process the action asynchronously in a background continuation step.

Serverless Cold Start Latency

First invocation after inactivity triggers a cold start that can add 1–2 seconds to the initial pipeline step. Mitigated by keeping serverless function bundles lean and relying on Vercel's automatic prewarming for high-traffic deployments.

LLM Output Consistency for Classification

LLM outputs are probabilistic — the same lead could produce different qualification categories on separate calls. Addressed by setting temperature=0 for all classification calls and using `generateObject` with a strict Zod enum schema to constrain outputs.

Research Quality Variance for Low-Footprint Companies

Exa.ai results vary significantly for early-stage or low-profile companies with minimal online presence. Addressed with fallback research prompts, a minimum word-count threshold check, and graceful degradation to Phase 1 data only.

14. Limitations

Product Limitations (v1.0)

Limitation	Description	Planned Fix
English-only	All prompts and email generation support English only	Multi-language support in v2.0
Single Slack channel	One notification channel per deployment	Per-category channel routing in v1.1
No CRM write-back	Leads not automatically created in CRM on qualification	Salesforce / HubSpot integration in v1.1
No lead deduplication	Duplicate form submissions create separate workflows	Deduplication by email in v1.1
No conversation history	Each lead treated in isolation — no prior interaction memory	CRM + history lookup in v1.1
No analytics dashboard	No web UI for qualification breakdown or pipeline trends	Analytics dashboard in v1.2
Stub service functions	sendEmail, crmSearch, queryKnowledgeBase are unimplemented placeholders	Must be implemented by adopting organisation before go-live

AI Model Limitations

- **Non-determinism** — LLMs may produce slightly different outputs for identical inputs; mitigated with temperature=0 and structured `generateObject`

- **Hallucination risk** — Research agent may synthesise plausible-sounding but incorrect facts; mandatory human Slack approval before email send prevents errors reaching prospects
- **Context window limits** — Very large research reports may require truncation for models with smaller context windows
- **Training data cutoff** — Model knowledge may not include the most recent events; Exa.ai real-time search compensates for this

Infrastructure Limitations

- **Serverless timeout** — Functions have a maximum execution time; long research steps must complete within Vercel Pro limits (~300s)
- **Vercel-only architecture** — v1.0 is designed for Vercel; self-hosting requires significant re-engineering
- **No data residency controls** — All data flows through cloud services; not suitable for strict GDPR on-premise requirements in v1.0

15. Assumptions

#	Assumption
A-01	The adopting team has an active Vercel Pro account with sufficient serverless execution time
A-02	A Slack workspace and bot with chat:write, actions, and event subscriptions permissions are configured
A-03	An Exa.ai API key is available with quota sufficient for the expected lead volume
A-04	The business has a clearly defined ICP that can be expressed in the qualification-rules.ts config file
A-05	SDRs are responsive on Slack during business hours to approve or reject email drafts in a timely manner
A-06	Inbound form submissions are treated as genuine in v1.0; spam protection is out of scope
A-07	All placeholder service stubs (sendEmail, crmSearch, queryKnowledgeBase) will be implemented before production go-live
A-08	The Vercel AI Gateway API key has access to the required LLM model for qualification and email generation

16. Release Roadmap & Milestones

v1.0 — MVP (Complete · 5-Week Development Sprint)

Week	Milestone	Status
Week 1	Project setup, Next.js 16 scaffold, contact form design, Zod validation schema	☑ Complete
Week 2	Phase 1 background verification — email, company, decision-maker, tech, financial, risk	☑ Complete
Week 3	AI deep research agent (Exa.ai), research report compiler, 4-category qualification engine	☑ Complete
Week 4	Email generation, Slack Bolt integration, human-in-loop approval and webhook handler	☑ Complete
Week 5	End-to-end testing, ICP config hardening, documentation, Vercel production deployment	☑ Complete

v1.1 — Enhanced Integrations (Target: Q2 2026)

Feature	Priority
CRM write-back — Salesforce / HubSpot auto lead creation	P0
Lead deduplication by email address	P0
Per-category Slack channel routing (SUPPORT → #support, QUALIFIED → #sales)	P1
Database persistence (Vercel Postgres / Neon) for lead history	P1
Dead-letter queue and retry logic for failed workflow steps	P1

v1.2 — Analytics & Observability (Target: Q3 2026)

Feature	Priority
Lead analytics dashboard — volume, categories, response time trends	P1
SDR performance metrics — approval rates, override rates per rep	P2
Anomaly detection alerts — spike in UNQUALIFIED rate above threshold	P1

v2.0 — Scale & Intelligence (Target: Q4 2026)

Feature	Priority
Multi-language support — auto-detect lead language, localise email	P1
A/B email template testing framework	P2
Custom qualification scoring model (org-specific fine-tuning)	P2
SDR override feedback loop feeding back into qualification rules	P1
Multi-channel notifications (Teams, Email, SMS alongside Slack)	P2

17. Open Questions

#	Question	Owner	Status
OQ-01	Which email sending service will be used? Resend, SendGrid, or AWS SES?	Engineering	Open
OQ-02	Which CRM is the primary integration target for v1.1?	Product / Sales	Open
OQ-03	Should UNQUALIFIED leads receive a polite decline email or be silently archived?	Sales Leadership	Open
OQ-04	What is the acceptable maximum pipeline latency under peak load?	Product	Open
OQ-05	Should the platform support multi-tenant Slack workspaces for a SaaS offering?	Product	Open
OQ-06	Are there GDPR or data-residency constraints for processing EU leads via Exa.ai?	Legal	Open
OQ-07	Which default LLM should be configured — GPT-4o, Claude 3.5 Sonnet, or other?	Engineering	Open
OQ-08	Should compiled research reports be stored and archived for future SDR review?	Product	Open

18. Appendix

A. Qualification Category Quick Reference

Category	Primary Trigger	Email Tone	SDR Action
QUALIFIED 	ICP match + decision-maker + urgency signals	Excited, personalised, clear CTA	Priority Slack review → Approve → Send
FOLLOW_UP 	Interest but wrong timing or information gap	Nurturing, supportive, keep door open	Review → Approve → Add to drip
UNQUALIFIED 	Red-flag keyword or ICP exclusion rule hit	Optional polite decline	Archive or dismiss in Slack
SUPPORT 	post-sale keyword (bug, error, login, refund)	Empathetic, support-routing	Route to support team — no SDR

B. ICP Configuration Reference

Config Field	Description	Default
minEmployees	Minimum company size for ICP fit	20
maxEmployees	Maximum company size	5,000
targetIndustries	Industries considered a strong fit	Software/SaaS, Technology, FinTech, Web3, E-commerce
minAnnualBudget	Minimum expected deal size	\$50,000
decisionMakerTitles	Job titles with purchasing authority	CTO, VP Engineering, Director of Engineering
targetRegions	Geographic regions served	North America, Europe, Asia-Pacific
excludedIndustries	Industries auto-classified UNQUALIFIED	Insurance, Government, Non-profit, Healthcare
excludeKeywords	Phrases triggering UNQUALIFIED	freelancer, solo developer, student, hobby project
supportKeywords	Phrases triggering SUPPORT	bug, error, broken, login, account issue, refund

C. Glossary

Term	Definition
ICP	Ideal Customer Profile — archetype of a company most likely to buy and succeed with the product
SDR	Sales Development Representative — reviews and acts on lead notifications in Slack
HIL	Human-in-the-Loop — the mandatory Slack approval step before any email reaches a prospect
Durable Workflow	Background job that survives server restarts and automatically retries failed steps
Exa.ai	Semantic web search API providing real-time search and URL content extraction
AI Gateway	Vercel's model-agnostic proxy for accessing multiple LLM providers via a single API key
generateObject	Vercel AI SDK function that returns structured JSON validated against a Zod schema
Workflow DevKit	Durable execution engine that runs the qualification pipeline as isolated, retryable steps

This PRD is a living document. Review and update at each sprint planning or business review.